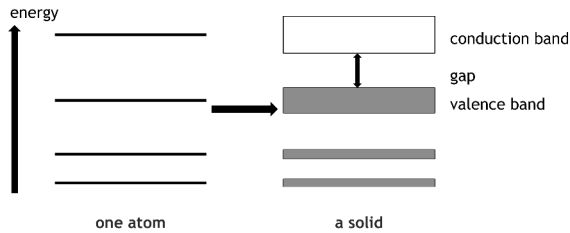
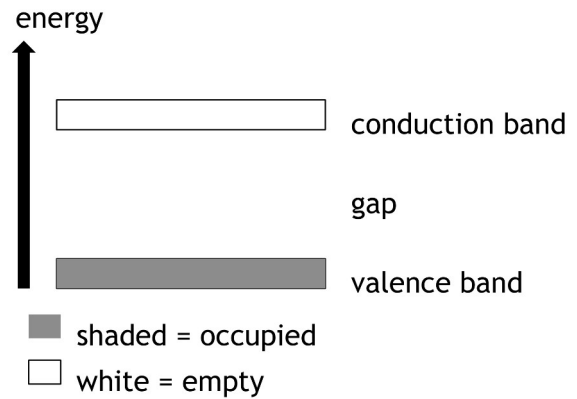


Cut out each diagram and stick it into your jotter under its heading. Shaded = occupied by electrons. White = empty.

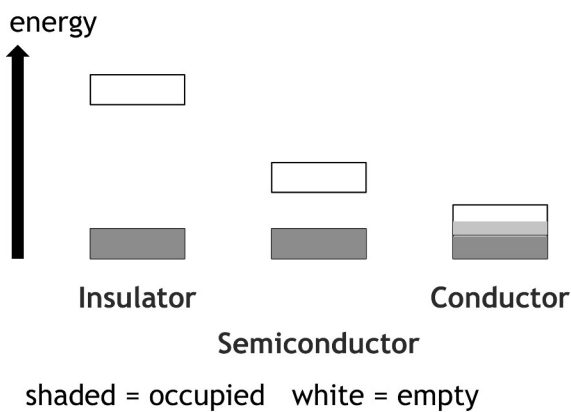
From energy levels to energy bands



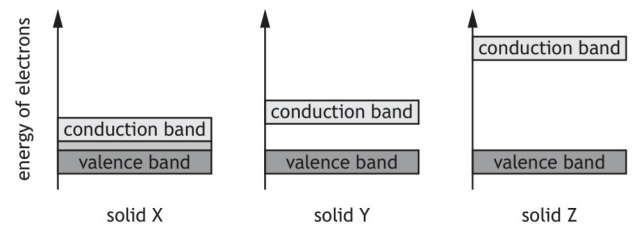
The conduction band and the valence band



Three kinds of solid

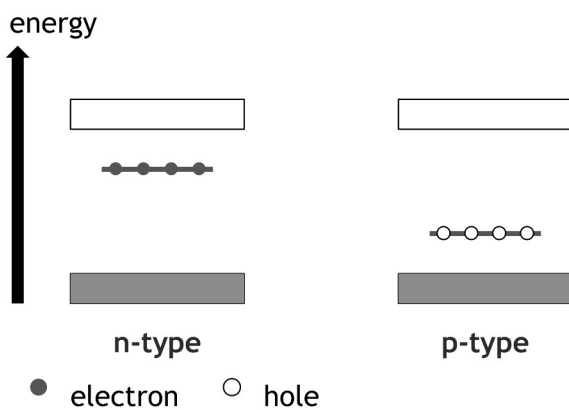


Three kinds of solid, as the exam draws it

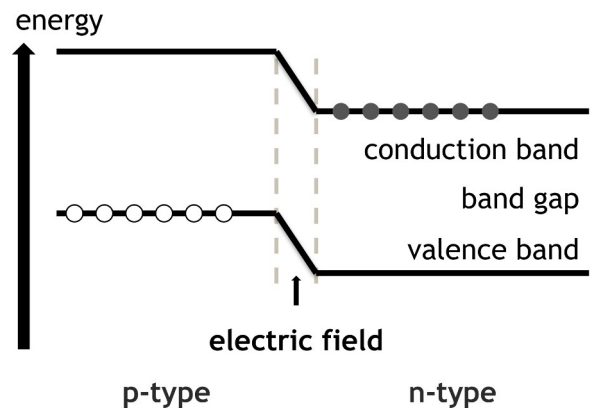


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Doping: n-type and p-type

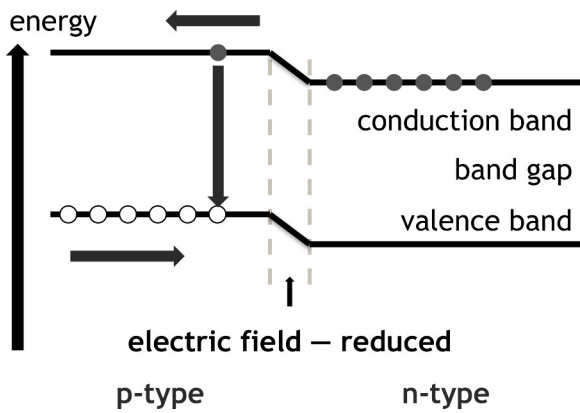


The p-n junction

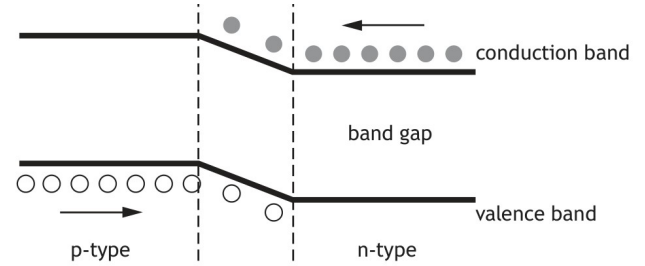


Cut out each diagram and stick it into your jotter under its heading. Shaded = occupied by electrons. White = empty.

Forward bias

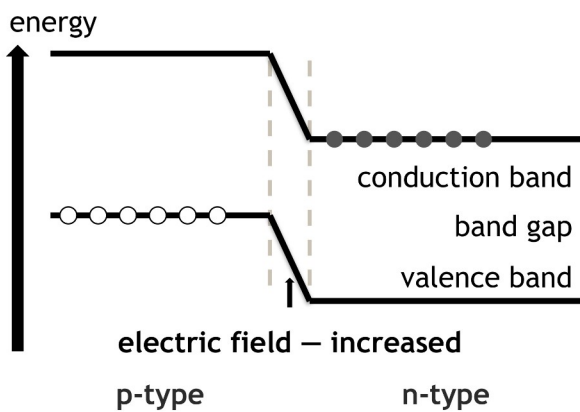


Forward bias, as the exam draws it



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Reverse bias



Spotting the bias in a circuit

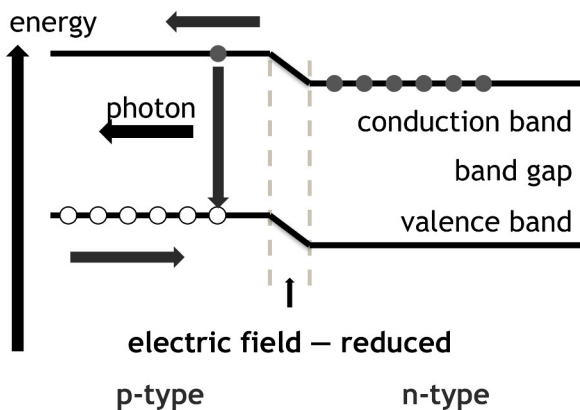


forward biased: conducts

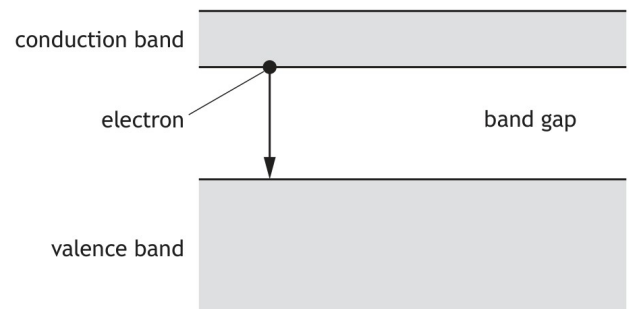


reverse biased: no current

How an LED emits light



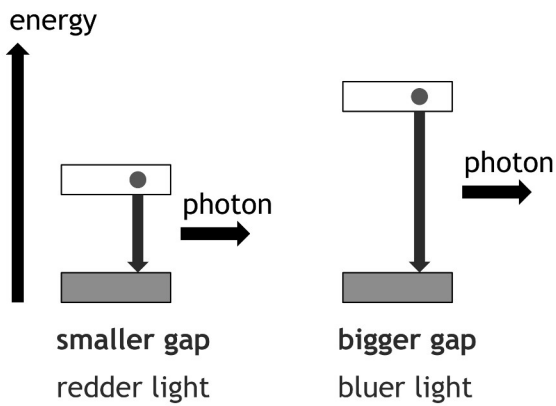
An LED's bands in an exam question



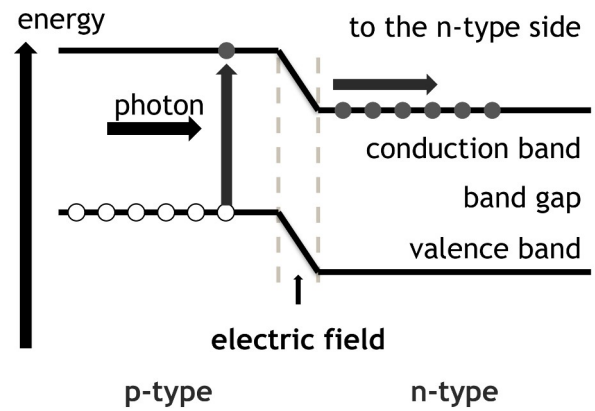
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Cut out each diagram and stick it into your jotter under its heading. Shaded = occupied by electrons. White = empty.

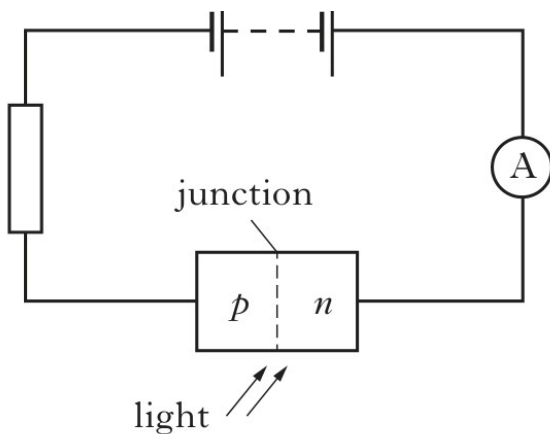
Band gap and colour



How a solar cell works



A p-n junction with light on it



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